

torch.segment_reduce#

https://pytorch.org/docs/stable/generated/torch.segment_reduce.html#torch.se

Description:

Perform a segment reduction operation on the input tensor along the specified axis.

- `data` (Tensor) – The input tensor on which the segment reduction operation will be performed.
- `reduce` (str) – The type of reduction operation. Supported values are `sum`, `mean`, `max`, `min`, `prod`.
- `lengths` (Tensor, optional) – Length of each segment. Default: `None`.
- `offsets` (Tensor, optional) – Offset of each segment. Default: `None`.

Function Signature

```
torch.segment_reduce(data: Tensor, reduce: str, *, lengths: Tensor | None = None, indices: Tensor | None = None, offsets: Tensor | None = None, axis: _int = 0, unsafe: _bool = False, initial: Number | _complex | None = None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

lengths (Tensor, optional) – Length of each segment. Default: None. offsets (Tensor, optional) – Offset of each segment. Default: None. axis (int, optional) – The axis perform reduction. Default: 0. unsafe (bool, optional) – Skip validation If True. Default: False. initial (Number, optional) – The initial value for the reduction operation. Default: None.

Code Examples

```
>>> data = torch.tensor([[1, 2, 3, 4],[5, 6, 7, 8],[9, 10, 11, 12]], dtype=torch.float32, device='cuda')
>>> lengths = torch.tensor([2, 1], device='cuda')
>>> torch.segment_reduce(data, 'max', lengths=lengths)
tensor([[ 5.,  6.,  7.,  8.],
        [ 9., 10., 11., 12.]], device='cuda:0')
```

Detailed Documentation

Documentation

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`data` (Tensor) – The input tensor on which the segment reduction operation will be performed.

`reduce` (str) – The type of reduction operation. Supported values are sum, mean, max, min, prod.

`lengths` (Tensor, optional) – Length of each segment. Default: None.

`offsets` (Tensor, optional) – Offset of each segment. Default: None.

`axis` (int, optional) – The axis perform reduction. Default: 0.

`unsafe` (bool, optional) – Skip validation If True. Default: False.

`initial` (Number, optional) – The initial value for the reduction operation. Default: None.